Learning When Black-Hole Hair Is Observable from Synthetic Ringdown Waves

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Abstract—We ask when a synthetic black-hole ringdown contains enough information to recover the Kiselev parameters (k, w_q), and when the inverse map is too ill-conditioned for finite-precision inference. To prevent repeated mode realizations of one spacetime from leaking across data partitions, we use grouped physical cross-validation and fit waveform PCA only on training folds. A waveform-only random forest reaches $R^2(k) = 0.8960$ and $R^2(w_q) = 0.9587$; adding synthetic geodesic proxies raises these scores to 0.9793 and 0.9891 with histogram gradient boosting. More importantly, an all-feature random-forest classifier reproduces a Fisher/Jacobian observability boundary with macro F1 = 0.9928, using $\sigma_{w_q} > 0.3$ at one-percent relative observable precision as the weak- identifiability criterion. In a companion scalar/proxy identifiability experiment, adding synthetic geodesic features reduces mean absolute w_q error from 0.0225 to 0.0143 and improves local conditioning near $k = 0$. The study is intended as a controlled scientific-ML benchmark rather than an observational analysis: the signals are synthetic leading-eikonal curves, and the geodesic features are replaceable proxies rather than ray-tracing outputs.

Index Terms—scientific machine learning, inverse problems, black-hole ringdown, identifiability, grouped validation, Fisher information

I. Introduction

This paper studies a specific inverse problem: recovering the Kiselev parameters (k, w_q) from the frequency and damping morphology of synthetic ringdown curves. In the eikonal limit, quasinormal frequencies are related to the orbital frequency and instability exponent of an unstable circular null geodesic [1], [2]. The relation gives a transparent simulator in which machine-learning behavior can be compared directly with the local derivatives of the observable map.

In this setting, a high R^2 can mean either genuine recovery of Kiselev parameters or interpolation across repeated mode realizations of the same spacetime. Each (k, w_q) point produces several curves through different angular indices ℓ and overtones n; a row-wise split can therefore place nearly identical physical systems in training and test sets. Even after that leakage is removed, a model may interpolate accurately in most of the grid while becoming unreliable where the inverse map is locally ill-conditioned. Our central question is consequently: when does a synthetic ringdown contain enough information to infer Kiselev hair parameters, and when are finite-precision errors amplified too strongly?

We address these issues in a synthetic Kiselev black-hole study. The Kiselev solution is a static, spherically symmetric geometry containing an amplitude k and state parameter w_q [3]. Our simulator uses leading-order analytic shifts rather than a dedicated perturbative solver. We generate damped curves, enforce grouped physical validation, fit PCA within training folds, and compare several inverse models. We then change the target: instead of predicting only (k, w_q), a classifier predicts whether w_q is practically identifiable under a stated precision model. A numerical Fisher/Jacobian calculation provides the scientific label and an analytic interpretation of failures.

We address that question through five connected components:

- grouped physical validation for waveform-to-hair inference, with PCA fitted inside each training fold;
- regression from synthetic leading-eikonal curves to (k, w_q);
- Fisher/Jacobian labels for weakly identifiable regions;
- comparison of waveform-only, scalar, and synthetic geodesic-proxy representations; and
- joint interpretation of ML errors and local singular-value diagnostics.

The central result is not the regression score by itself. The more useful outcome is that a learned classifier reproduces the Fisher/Jacobian observability boundary, separating regions where the waveform-to-hair map is informative from regions where finite-precision errors are amplified.

II. Synthetic Leading-Eikonal Waveform Dataset

A. Static Kiselev simulator

We set the mass scale to $M = 1$ and sample a dense grid

$$k \in [-0.04, 0.04], \qquad w_q \in [-1.4, 1.4]. \tag{1}$$

The underlying Kiselev metric family motivates the dependence on both parameters [3]; the present implementation uses analytic leading-order expressions for the photon-orbit radius shift Δr, orbital frequency Ω, and Lyapunov exponent λ. Non-finite or non-positive leading-order points are excluded. The broad w_q interval is used as a controlled benchmark of inverse-map geometry and is not intended to identify every sampled point with the standard quintessence regime.

For each retained physical point, the eikonal mapping is

$$\omega_R = \ell\Omega, \qquad \gamma = (n + \tfrac{1}{2})\lambda, \tag{2}$$

and the synthetic time-domain curve is

$$\Psi(t) = \exp(-\gamma t)\cos(\omega_R t). \tag{3}$$

We use $\ell \in \{4, 5, 10, 20\}$, $n \in \{0, 1\}$, and 512 uniformly spaced time samples. The eight mode curves are averaged into one 512-sample multi-mode morphology per physical point for the waveform-to-hair experiment. This aggregation prevents repeated modes from being treated as independent target samples while retaining their combined morphology. The equal-weight average is a benchmark feature construction rather than an astrophysical multimode waveform with physically inferred amplitudes and phases.

B. Feature families

Five representations are evaluated: (A) the raw waveform; (B) waveform PCA; (C) waveform PCA with (Ω, λ); (D) waveform PCA with four synthetic geodesic proxies; and (E) waveform PCA with all current scalars and proxies. The current scalars are Ω, λ, ω_R, γ, and Δr.

The additional features are smooth proxies for impact parameter, screen coordinate, propagation-time delay, and a redshift curve. They exercise a replaceable multi-observable interface and test whether parameter dependence not reducible to (Ω, λ) can improve conditioning. They should not be interpreted as geodesic-shooting or ray-tracing predictions.

III. Leakage-Aware Inverse-Learning Protocol

A. Grouped physical validation

The grouping variable is the physical pair (k, w_q); neither ℓ nor n defines a new physical system. Grouped cross-validation therefore assigns all mode realizations of one physical point to exactly one fold. This is the central leakage control. For the zero-noise headline experiment, PCA is fitted on clean training curves. In the noise study, independent noise is added to training and test curves, and PCA is fitted only on the noisy training fold at each noise level. PCA never sees held-out test data. This follows the general train-transform-test discipline implemented with scikit-learn [4].

Regression candidates are Ridge, random forest, histogram gradient boosting, and a multilayer perceptron. Classification candidates are logistic regression, random forest, histogram gradient boosting, and a multilayer perceptron. Random forests provide a robust nonlinear baseline [5]; gradient boosting supplies a complementary additive-tree model [6]. Hyperparameters are deliberately modest because the scientific question concerns information content rather than model scaling.

Regression is measured using R^2, mean absolute error (MAE), and root mean squared error for each target. Classification is measured with accuracy, macro F1, and a confusion matrix. Gaussian waveform noise at 0, 0.5, 1, 3, and 5 percent of the per-waveform standard deviation is injected before PCA or direct waveform input.

B. Practical-identifiability label

Let the local observable vector be

$$\mathbf{y}(k, w_q) = (\Omega, \lambda, \omega_R, \gamma, \Delta r)^{\mathsf{T}}. \tag{4}$$

Finite differences give the Jacobian

$$J = \frac{\partial \mathbf{y}}{\partial(k, w_q)}. \tag{5}$$

Under independent one-percent relative errors, with a small absolute floor for zero-valued shifts, we whiten the Jacobian as

$$\widetilde{J} = C_y^{-1/2} J = U \operatorname{diag}(s_i)\, V^{\mathsf{T}}. \tag{6}$$

Singular values below $\max(\epsilon_{\rm abs}, \epsilon_{\rm rel} s_{\max})$, with $\epsilon_{\rm rel} = 10^{-8}$ and $\epsilon_{\rm abs} = 0$, define the numerical null space. This tolerance is supported by derivative convergence: five-point and three-point whitened Jacobians differ relatively by at most 9.31×10^{-9} at representative near-degenerate points, and scaling both finite-difference steps from 0.25 to 4 leaves all labels unchanged. If the explicit w_q direction overlaps the numerical null space, we set $\sigma_{w_q} = \infty$ and mark the parameter structurally or numerically non-identifiable. Otherwise the covariance is assembled from retained singular directions, $\Sigma_\theta = V_r \operatorname{diag}(s_r^{-2}) V_r^{\mathsf{T}}$. This avoids the incorrect finite zero uncertainty that a bare pseudoinverse assigns to a discarded parameter direction.

At exactly $k = 0$, the observables have no w_q dependence, the Jacobian has rank one, and $\sigma_{w_q} = \infty$. At small nonzero $|k|$, by contrast, the Jacobian can remain full rank while producing a large but finite uncertainty. We call this practical weak identifiability, labeling a point weak when

$$\sigma_{w_q} = \sqrt{(\Sigma_\theta)_{w_q w_q}} > 0.3. \tag{7}$$

This criterion labels 57.6% of proxy-valid physical points as weakly identifiable. It is a simulator-specific, precision-dependent phase diagram, not an observational posterior.

C. Reproducibility and audit trail

The experiment is implemented in the open research prototype black-hole-hair-ml [7]. Configuration files record the dense parameter ranges, mode indices, number of time samples, PCA dimension, model settings, and noise levels. Each run exports fold-level metrics, out-of-fold predictions, noise tables, paired PNG/PDF figures, and an automatically generated scope-aware report. The physical group identifier is stored explicitly with predictions, allowing a post-run check that no group crosses the train/test boundary. The PCA object is constructed inside the fold loop rather than fitted once on the full dataset. These implementation details are important because a numerically strong result would not be scientifically interpretable

TABLE I

Grouped waveform-to-hair results. Geodesic features are synthetic proxies, not physical ray-tracing outputs.

Input and best regressor	$R^2(k)$	$R^2(w_q)$
Waveform only, random forest	0.8960	0.9587
Waveform + proxies, HistGB	0.9793	0.9891

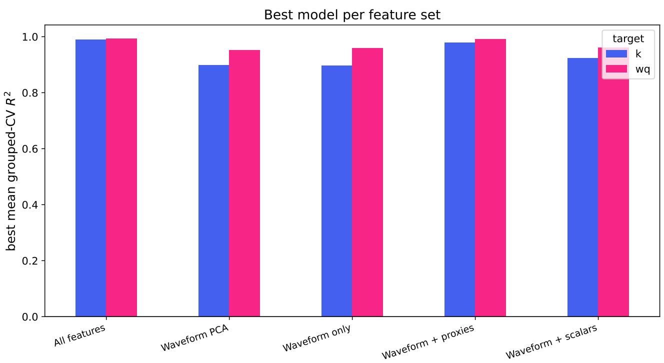

Fig. 1. Grouped-CV feature comparison using the best model for each feature-set/target pair. "Proxies" denotes synthetic geodesic features rather than physical shooting or ray tracing.

if preprocessing or repeated-mode leakage exposed test information during training.

The present workshop draft reports the completed default experiment rather than a hyperparameter optimization study. Model families were compared under the same grouped partitions, and the best named estimators are selected from that controlled comparison. The feature comparison reports the best mean grouped-CV score for each feature-set/target pair.

IV. Waveform-to-Hair Prediction Results

Table I summarizes the main grouped results. Waveform morphology alone is informative: the random forest reaches $R^2(k) = 0.8960$ and $R^2(w_q) = 0.9587$. Because Eq. (3) is fully determined by $(\Omega, \lambda, \ell, n)$, this performance measures how well the model inverts the frequency and damping morphology within the sampled simulator. It does not demonstrate additional physical information in the curve.

Adding proxy observables raises the jointly selected model scores to 0.9793 and 0.9891 with histogram gradient boosting. Figure 1 gives the complementary best-per-target comparison across representations. Within this simulator, additional parameter dependence improves inverse recovery; the next physics question is whether validated observables retain that advantage.

The optional true-versus-predicted diagnostic in Fig. 2 shows the held-out structure for the best all-feature model. Residuals are not uniform over parameter space; this motivates the observability-classification task rather than treating a global score as a complete scientific result.

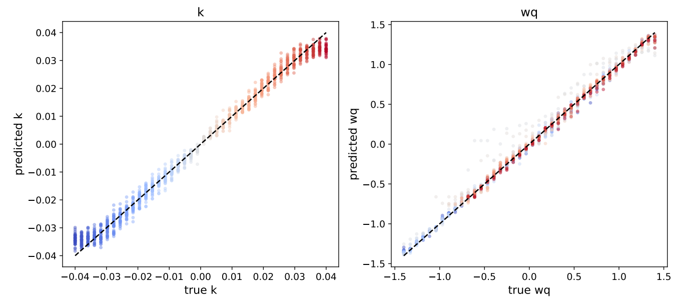

Fig. 2. True versus grouped-CV predicted k and w_q for the all-feature histogram-gradient-boosting model, including waveform PCA, scalars, and synthetic geodesic proxies.

Waveform-only w_q MAE rises by more than 25% at the first tested nonzero noise level, 0.5%. This sensitivity should not be interpreted as a detector threshold: the perturbation is idealized Gaussian noise on normalized simulator curves and omits calibration, colored noise, mode uncertainty, start-time selection, and waveform systematics.

V. Learning Observability and Analytic Identifiability

Figure 3 compares the analytic class with predictions from the best all-feature random-forest classifier. Its macro F1 is 0.9928. The model captures both the broad transition in w_q and the narrow structure around small $|k|$. This is the central result: the classifier recovers where the inverse problem is informative and where finite-precision uncertainty is strongly amplified, rather than merely returning a parameter estimate.

An independently evaluated $k = 0$ limit provides a structural sanity check: the metric reduces to Schwarzschild and w_q becomes unobservable. The nontrivial result shown by the sampled grid is the finite-width weak-identifiability region at small nonzero $|k|$.

A companion proxy-identifiability experiment provides a complementary view of this boundary in Fig. 4. Large condition number marks amplification of observable perturbations in the inverse problem, while the minimum singular value measures the weakest local parameter direction. In that companion scalar/proxy experiment, proxy augmentation reduces mean absolute w_q error from 0.0225 to 0.0143. In a near-$k = 0$ band, the median minimum singular value increases by approximately $12\times$, and the median condition number improves by approximately $2.13\times$. Improvement at small nonzero k does not remove the independently verified exact-$k = 0$ rank loss.

The agreement between learned classification and analytic sensitivity is more scientifically informative than the regression score alone. It distinguishes model failure from an inverse map whose local geometry suppresses information. At the same time, the comparison remains internal to the assumed formulas and error model. A different observable covariance or a validated perturbative spectrum would move the practical boundary.

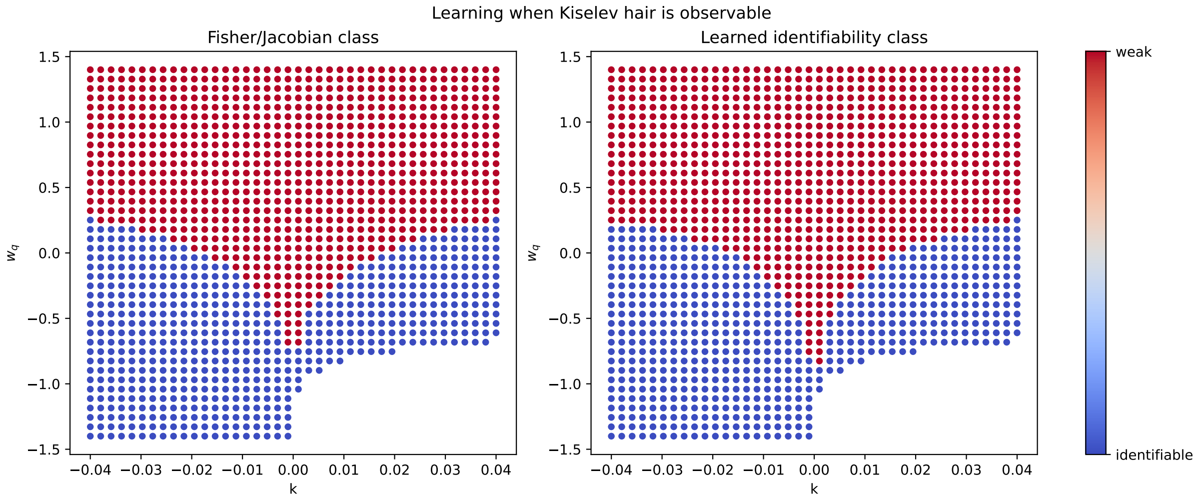

Fig. 3. True Fisher/Jacobian identifiability class (left) and grouped-CV prediction (right). Weak identifiability means $\sigma_{w_q} > 0.3$ under 1% relative observable precision. The all-feature random-forest classifier reaches macro F1 = 0.9928. The independently verified $k = 0$ limit is Schwarzschild; the nontrivial result shown here is the learned finite-precision boundary at small nonzero $|k|$.

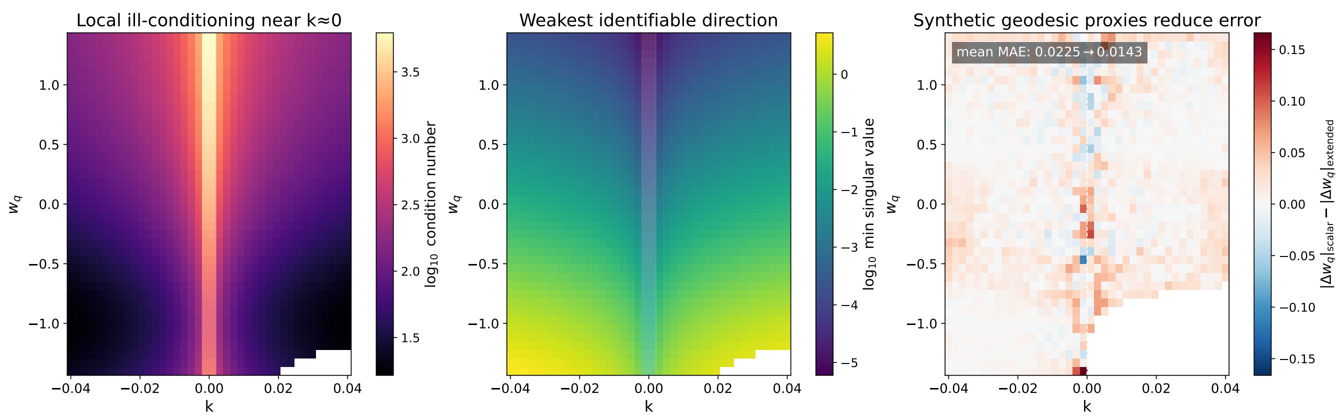

Fig. 4. Analytic identifiability lens. Left: Jacobian condition number. Center: minimum singular value. Right: reduction in w_q error after adding synthetic geodesic proxies. The proxies are not physical ray-tracing or shooting outputs.

VI. Limitations and Future Work

Synthetic waveform scope. The inputs are generated by Eq. (3), not measured strain. Detector response, colored noise, calibration error, mode mixing, and ringdown start-time uncertainty are absent, so the noise experiment is a robustness test rather than an observational sensitivity forecast. The mode-averaged morphology is an equal-weight feature construction rather than an astrophysical multimode waveform with physically inferred amplitudes and phases.

Leading-eikonal/QNM scope. The geodesic–QNM correspondence is a leading-order approximation [1]. Dedicated perturbative spectra are needed to determine whether the same observability structure survives finite-ℓ corrections and waveform modeling error.

Proxy-observable scope. The impact-parameter, screen-coordinate, propagation-time, and redshift quantities are smooth synthetic proxies. Their improvement demonstrates the value of new functional dependence, not the performance of physical ray tracing or a proposed observing program.

Validation and extrapolation scope. Grouped CV removes mode-level leakage but primarily measures interpolation across the sampled family. Blocked parameter tests, repeated sampling designs, calibrated predictive uncertainty, and simulator-mismatch studies are required before making claims about extrapolation.

The next physics step is to replace proxies with converged direct and secondary geodesic-shooting outputs and recompute the enlarged Jacobian. The next waveform step is validation against dedicated Kiselev perturbative calculations, including multiple modes and finite-ℓ corrections. Only after these checks should the framework be connected to public-event posteriors or a non-GR strain-level likelihood.

VII. Conclusion

We presented a leakage-aware scientific-ML study of Kiselev waveform-to-hair inference. Grouping by physical parameters and fitting PCA only on training folds provide a credible interpolation test. Synthetic waveform morphology recovers (k, w_q) well inside the assumed simulator, while synthetic geodesic-proxy augmentation further improves regression. More importantly, an observability classifier learns a Fisher/Jacobian phase boundary with macro F1 0.9928, linking machine-learning errors to the local geometry of the inverse problem. The independently evaluated $k = 0$ limit establishes structural non-identifiability, while the sampled small-nonzero-$|k|$ region reveals full-rank but practically weak identifiability. This suggests a general AI-for-science principle: inverse models should learn when parameters are observable, not merely fit them.

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